

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.* (BTL 3)

Assessment Tool Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I K Sathwik Kumar (192425051) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Sathwik Kumar

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	

8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

Q.No	Use Case Scenario	
1	Smart Healthcare Diagnosis	A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention. Tasks: (i) Identify facts, entities, predicates, variables, and constants. (ii) Represent the knowledge using Propositional Logic. (iii) Represent the knowledge using FOL with suitable quantifiers. (iv) Perform unification for two relevant predicates. (v) Apply resolution to prove whether Ravi has a viral infection. (vi) Show the complete inference sequence and final conclusion.

QUESTION 1 – SMART HEALTHCARE DIAGNOSIS

Scenario

A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention.

(i) Facts, Entities, Predicates, Variables and Constants

1. Entity

- **Ravi** – the patient being diagnosed.

2. Constants

- Ravi is a constant representing the specific patient.

3. Facts

The known facts are:

- Fever(Ravi)
- Cough(Ravi)
- BodyPain(Ravi)

4. Predicates

Predicate	Meaning
Patient(x)	x is a patient
Fever(x)	x has fever
Cough(x)	x has cough
BodyPain(x)	x has body pain
Flu(x)	x has flu
VirallInfection(x)	x has a viral infection
MedicalAttention(x)	x requires medical attention

5. Variable

- x represents any patient.

6. Relationships / Rules

1. Fever and cough → Flu
2. Flu and body pain → Viral infection
3. Viral infection → Medical attention

These are the main knowledge components that need to be reasoned over.

(ii) Propositional Logic Representation

Let:

- F = Ravi has fever
- C = Ravi has cough
- B = Ravi has body pain
- L = Ravi has flu
- V = Ravi has viral infection
- M = Ravi requires medical attention

Facts

[
F
]
[
C
]

[
B
]

Rules

Rule 1:

If Ravi has fever and cough, then Ravi has flu.

[
(F $\wedge$ C) $\rightarrow$ L
]

Rule 2:

If Ravi has flu and body pain, then Ravi has viral infection.

[
(L $\wedge$ B) $\rightarrow$ V
]

Rule 3:

If Ravi has viral infection, then Ravi requires medical attention.

[
V $\rightarrow$ M
]

Complete propositional knowledge base

[
F $\wedge$ C $\wedge$ B
]

[
(F $\wedge$ C) $\rightarrow$ L
]

[
(L $\wedge$ B) $\rightarrow$ V
]

[
V $\rightarrow$ M
]

The required query is:

[
 $\boxed{V}$
]

Therefore, we need to determine whether ViralInfection(Ravi) can be logically derived.

(iii) First-Order Logic Representation

Predicates

- Fever(x)
- Cough(x)
- BodyPain(x)
- Flu(x)
- ViralInfection(x)
- MedicalAttention(x)

Facts about Ravi

[
Fever(Ravi)
]

[
Cough(Ravi)
]

[
BodyPain(Ravi)
]

Rule 1

All patients who have fever and cough may have flu:

[
 $\boxed{\forall x[(Fever(x) \wedge Cough(x)) \rightarrow Flu(x)]}$
]

Rule 2

All patients who have flu and body pain may have viral infection:

[
 $\boxed{\forall x[(Flu(x) \wedge BodyPain(x)) \rightarrow ViralInfection(x)]}$
]

Rule 3

All patients with viral infection require medical attention:

[
 $\boxed{\forall x[ViralInfection(x) \rightarrow MedicalAttention(x)]}$
]

Existential representation

The scenario also establishes that at least one patient exists with the given symptoms:

[
 $\boxed{\exists x[Patient(x) \wedge Fever(x) \wedge Cough(x) \wedge BodyPain(x)]}$
]

For this scenario, that individual is Ravi.

Quantifiers Used

- $\forall$ = Universal quantifier – applies a rule to every relevant patient.
- $\exists$ = Existential quantifier – indicates that at least one patient with the stated properties exists.

(iv) Unification and Substitution

Unification makes two logical expressions identical by finding suitable substitutions.

Consider:

[
Fever(x)
]

and

[
Fever(Ravi)
]

For these expressions to match:

[
 $\boxed{x/Ravi}$
]

Therefore, the Most General Unifier (MGU) is:

[
 $\boxed{\theta = \{x/Ravi\}}$
]

After substitution:

[
 $Fever(x) \theta = Fever(Ravi)$
]

Similarly:

[
 $Flu(x)$
]
and
[
 $Flu(Ravi)$
]

can be unified using:

[
 $\theta = \{x/Ravi\}$
]

So the same patient variable x is substituted with the constant Ravi.

(v) Resolution to Prove Ravi Has Viral Infection

We want to prove:

[
 $ViralInfection(Ravi)$
]

Using resolution by refutation, we assume the opposite:

[
 $\neg ViralInfection(Ravi)$
]

Step 1 – Convert Rule 1 to clause form

Original:

[
 $(Fever(x) \wedge Cough(x)) \rightarrow Flu(x)$
]

Equivalent clause:

[
 $\neg Fever(x) \vee \neg Cough(x) \vee Flu(x)$
]

After substituting x = Ravi:

[
 $\neg Fever(Ravi) \vee \neg Cough(Ravi) \vee Flu(Ravi)$
]

We have:

[
 $Fever(Ravi)$
]
and
[
 $Cough(Ravi)$
]

Therefore:

[
 $\boxed{\text{Flu}(\text{Ravi})}$
]

Step 2 – Convert Rule 2 to clause form

Original:

[
 $(\text{Flu}(x) \wedge \text{BodyPain}(x)) \rightarrow \text{ViralInfection}(x)$
]

Clause form:

[
 $\boxed{\neg \text{Flu}(x) \vee \neg \text{BodyPain}(x) \vee \text{ViralInfection}(x)}$
]

Substitute:

[
 $x = \text{Ravi}$
]

giving:

[
 $\neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi}) \vee \text{ViralInfection}(\text{Ravi})$
]

Step 3 – Add the negated query

Query:

[
 $\text{ViralInfection}(\text{Ravi})$
]

Negated query:

[
 $\boxed{\neg \text{ViralInfection}(\text{Ravi})}$
]

Resolve:

[
 $\neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi}) \vee \text{ViralInfection}(\text{Ravi})$
]

with:

[
 $\neg \text{ViralInfection}(\text{Ravi})$
]

Result:

[
 $\boxed{\neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi})}$
]

Step 4 – Use the derived fact $\text{Flu}(\text{Ravi})$

We already derived:

[
 $\text{Flu}(\text{Ravi})$
]

Resolve:

[
 $\neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi})$
]

with:

[
 $\text{Flu}(\text{Ravi})$
]

Result:

[
 $\boxed{\neg \text{BodyPain}(\text{Ravi})}$
]

Step 5 – Use the known fact BodyPain(Ravi)

Known fact:

[
 $\text{BodyPain}(\text{Ravi})$
]

Resolve:

[
 $\neg \text{BodyPain}(\text{Ravi})$
]

with:

[
 $\text{BodyPain}(\text{Ravi})$
]

This produces:

[
 $\boxed{\Box}$
]

where $\Box$ represents the **empty clause / contradiction**.

Therefore, the assumption:

[
 $\neg \text{ViralInfection}(\text{Ravi})$
]

is false.

Component	Details
Entity	Ravi
Constants	Ravi
Variables	x
Facts	Fever(Ravi), Cough(Ravi), BodyPain(Ravi)
Predicates	Fever(x), Cough(x), BodyPain(x), Flu(x), ViralInfection(x), MedicalAttention(x)
Query	ViralInfection(Ravi)

(vi) Complete Inference Sequence

The complete reasoning chain is:

```
[  
Fever(Ravi)+Cough(Ravi)  
]
```

↓

```
[  
Flu(Ravi)  
]
```

↓

```
[  
Flu(Ravi)+BodyPain(Ravi)  
]
```

↓

```
[  
ViralInfection(Ravi)  
]
```

↓

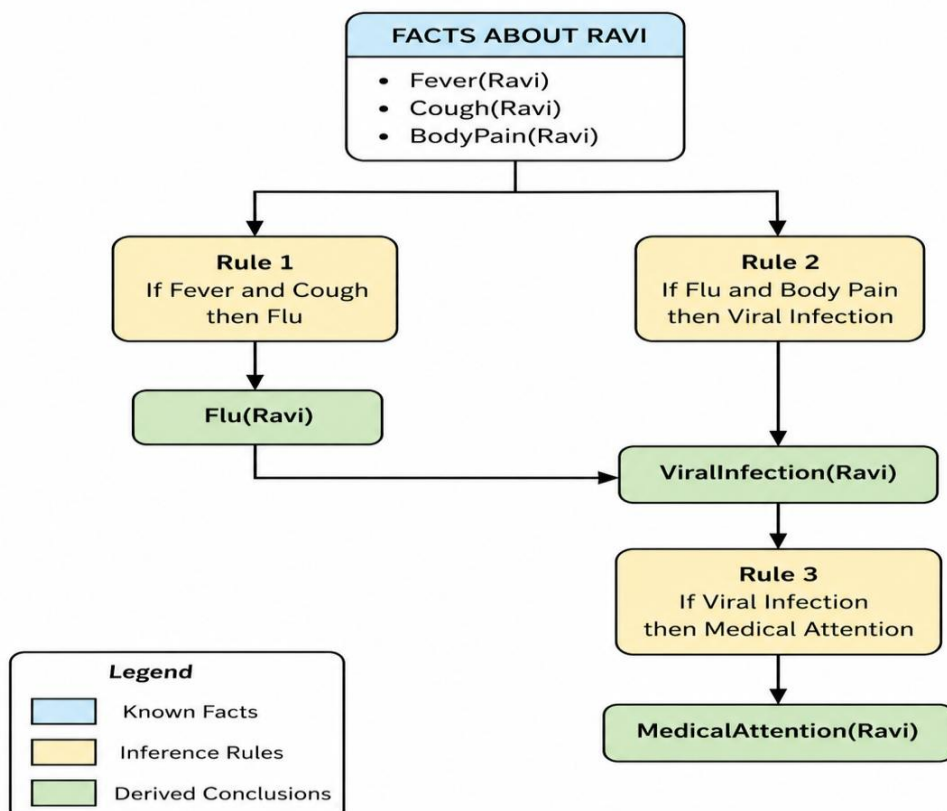
```
[  
MedicalAttention(Ravi)  
]
```

Final Conclusion

```
[  
\boxed{\text{Ravi has a viral infection and requires medical attention.}}  
]
```

This conclusion follows directly from the facts and rules provided in the scenario.

Knowledge Mapping / Logic Diagram



EVALUATION RUBRICS:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% Identifies most relevant knowledge with minor omissions.	7–9% Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% Provides mostly correct logical mappings with minor errors.	9–12% Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants & Quantifiers	15%	13–15% Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% Uses the concepts correctly with minor errors.	7–9% Demonstrates partial understanding with some incorrect usage.	0–6% Concepts are incorrectly applied or missing.

Logical Connectives & Formula Construction	10%	9–10% Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.
		construct valid logical formulas.			
Inference & Logical Reasoning	15%	13–15% Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% Reasoning is mostly correct with minor gaps.	7–9% Provides partial reasoning with some incorrect conclusions.	0–6% Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% Correctly performs most unification steps with minor errors.	5–6% Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% Resolution is incorrect or not demonstrated.

Clarity, Mapping Diagram Presentation	& 5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.
---------------------------------------	------	--	---	---	---